

Clinical Trial Executive Report

Drug X vs. Placebo — Phase III Efficacy & Safety Analysis

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1. Trial Overview & Demographics

Metric	Value
Total patients enrolled	500
Drug X arm	250
Placebo arm	250
Mean age (range)	54.5 (30-75)
Female patients	53.2%
Trial completion rate	84.8%
Dropout rate	15.2%

2. Primary & Secondary Endpoint Results

On average, the Drug X arm scored 7.90 points higher than Placebo on primary outcome score (0-100) (60.05 vs 52.15). This difference is statistically significant (Welch's $t=7.98$, $p<0.001$), meaning we would see a difference this large by chance in fewer than 0.1% of repeated trials. The effect size is medium (Cohen's $d=0.71$).

A chi-square test of independence was run on HbA1c <7% control at Week 24 across treatment arms ($\chi^2=56.11$, $\text{dof}=1$, $p<0.001$). There is a statistically significant association between treatment arm and HbA1c <7% control at Week 24.

3. Safety Profile

A chi-square test of independence was run on adverse event occurrence across treatment arms ($\chi^2=43.35$, $\text{dof}=1$, $p<0.001$). There is a statistically significant association between treatment arm and adverse event occurrence.

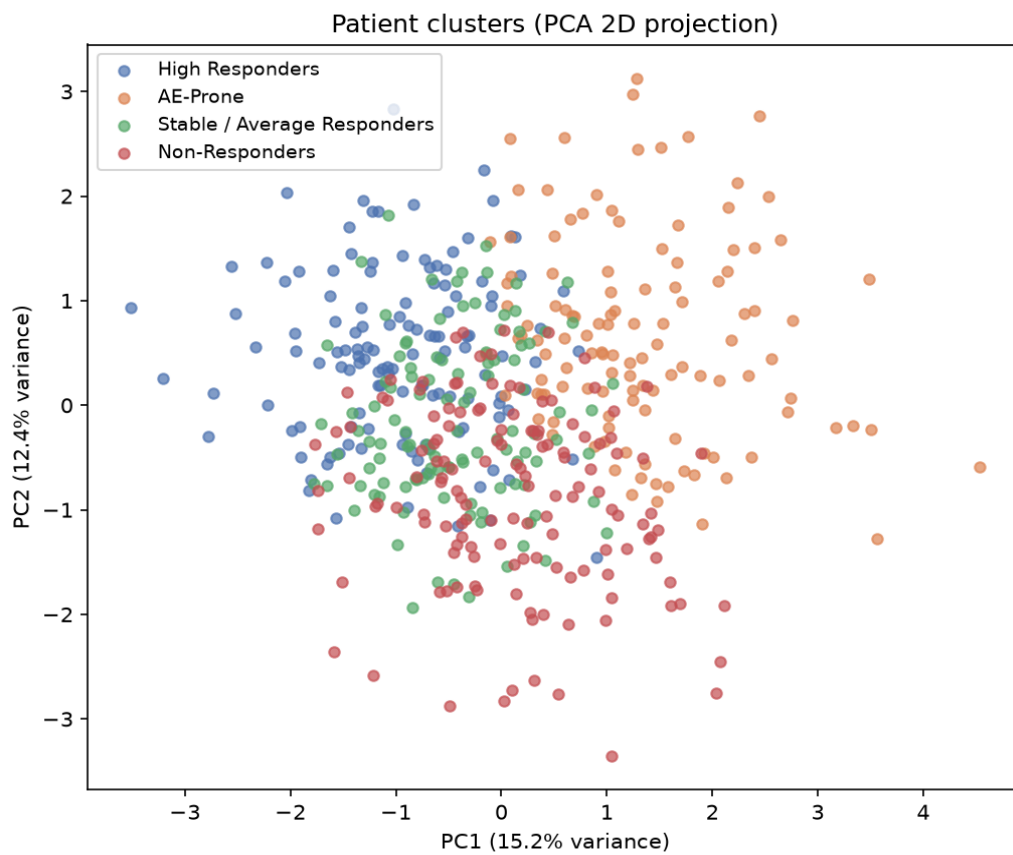
A chi-square test of independence was run on adverse event severity class across treatment arms ($\chi^2=49.90$, $\text{dof}=2$, $p<0.001$). There is a statistically significant association between treatment arm and adverse event severity class.

A Kruskal-Wallis test compared HbA1c reduction by Week 24, grouped by worst AE severity across 3 groups without assuming normally distributed data ($H=35.41$, $p<0.001$). The groups differ significantly in HbA1c reduction by Week 24, grouped by worst AE severity. Median values: No AE=0.32, Mild AE=0.67, Severe AE=0.44.

4. Patient Cluster Profiles (K-Means Segmentation)

K-Means with k=4 clusters (silhouette score 0.0894).

Cluster	N	Label	Outcome Score	Any-AE Rate	Severe-AE Rate	Dropout Rate
0	129	High Responders	61.535	49.6%	0.0%	7.8%
1	120	AE-Prone	48.293	99.2%	38.3%	40.8%
2	112	Stable / Average Responders	58.538	60.7%	0.0%	7.1%
3	139	Non-Responders	55.832	46.8%	0.0%	6.5%

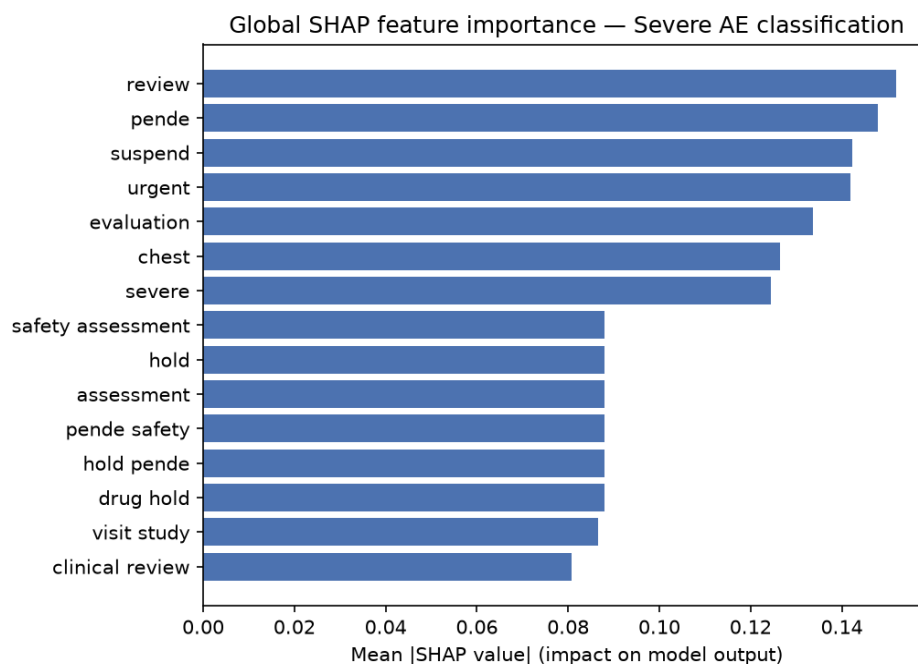


5. NLP Adverse-Event Classifier & Explainability

Metric	Value
Accuracy	92.2%
Precision (macro)	78.6%
Recall (macro)	94.9%
F1 (macro)	0.851

On the held-out test set, the NLP classifier detected 131 adverse-event notes vs. 73 caught by manual CRF coding (79.5% uplift), and recovered 100.0% of the 19 adverse events manual coding missed entirely.

SHAP global feature importance — Severe AE classification:



6. Top Clinical Insights

The following insights and recommendations are generated by Agent 3 (Insight Generation Agent), a live LLM call grounded in the statistics, clustering, and classifier tool results above.

1. The trial's primary endpoint shows a statistically significant difference between the Drug X and Placebo arms, with an average score 7.90 points higher for Drug X (Welch's $t=7.98$, $p<0.001$).
2. The AE classifier achieved an accuracy of 0.9217 and a weighted F1 score of 0.9277, indicating strong performance in detecting adverse events.
3. Cluster analysis revealed four distinct patient profiles, including "High Responders" with a mean outcome score of 61.535 and "AE-Prone" patients with a severe AE rate of 0.383.
4. The Kruskal-Wallis test showed a significant difference in HbA1c reduction by Week 24 across AE severity groups ($H=35.41$, $p<0.001$), with median HbA1c reductions of 0.32, 0.67, and 0.44 for No AE, Mild AE, and Severe AE groups, respectively.
5. The business impact analysis of the AE classifier showed an uplift of 79.5% in detected AEs compared to manual coding, with a recovery rate of 100.0% of manually missed AEs.

7. Recommendations for Next Steps

1. Based on the significant difference in primary endpoint scores, Drug X should be considered for further development and potential market approval.
2. The AE classifier's high accuracy and F1 score suggest its potential for use in future trials to improve AE detection and reduce manual coding errors.
3. The cluster analysis results should inform patient stratification and personalized treatment strategies in future trials, particularly for "AE-Prone" patients who may require closer monitoring.